



## F1-20 Osborne Reynolds' Demonstration

This item is intended to reproduce the classic experiments conducted by Professor Osborne Reynolds concerning the nature of laminar and turbulent flow.

- Reproducing the classic experiments conducted by Professor Osborne Reynolds concerning fluid flow condition
- Observing the laminar, transitional, turbulent flow and velocity profile

**PRODUCT CODE: F1-20**

**Categories:** Educational, F1 Series Fluid Mechanics, F1-20 - Osborne Reynolds Demonstration

**Tags:** Armfield , Dye Injection , Elementary Level , Flow Channel , Flow Visualisation , Fluid Mechanics , Hydraulics Bench , Laminar Flow , Open Channel Flow , Osborne Reynolds Demonstration , Transitional Flow , Turbulent Flow , Velocity Profile

### DESCRIPTION

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The equipment operates in a vertical orientation. A header tank containing stilling media provides a constant head of water through a bellmouth entry to the flow visualisation pipe.

Flow through this pipe is regulated using a control valve at the discharge end. The flow rate of water through the pipe can be read on the digital flow meter or can be measured using the volumetric tank (or measuring cylinder) of the hydraulics bench. Velocity of the water can therefore be determined to enable calculation of Reynolds' number.

The equipment uses a similar dye injection technique to that of Reynolds' original apparatus to enable observation of flow conditions.

### TECHNICAL SPECIFICATIONS

**Test pipe diameter:** 10.0mm (precision bore glass)

**Length of test pipe:** 700mm

**Dye reservoir capacity:** 0.45l

Requires Hydraulics Bench Service unit F1-10/F1-10-2

## FEATURES & BENEFITS

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### Experimental content

- Reproducing the classic experiments conducted by Professor Osborne Reynolds concerning fluid flow condition
- Observing the laminar, transitional, turbulent flow and velocity profile

### Dimensions

**Length:** 0.50m

**Width:** 0.33m

**Height:** 1.24m

### Order Codes

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